

Stellar Interiors, Nucleosynthesis & Evolution

stars have midlife crises too, they also curl up into a ball and cry

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Prerequisite Knowledge

- The ideal gas law
- Basic thermodynamics of heat transport
- Atomic mass transformations
- Single-variable calculus
- The Pauli exclusion principle
- Braincells, now under enormous internal pressure

1 Introduction

Imagine you're a star. You think you're just sitting up there looking pretty, but you're actually trapped in a brutal, never-ending tug-of-war.

On one side, you've got gravity. It's incredibly heavy, completely relentless, and it has one single-minded goal: crush everything you are down into a tiny, microscopic dot.

But you don't cave. Why? Because you've got a massive internal defense squad pushing back out—gas pressure, radiation, and eventually, the weird rules of quantum mechanics that refuse to let matter be squeezed any further.

When these two forces completely lock horns and nobody is winning, you're in a perfect stalemate. Physicists call this the "main sequence." To you, it just means you get to sit back, relax, and quietly burn through your hydrogen for billions or trillions of years.

Every major, dramatic event in your life—your birth, your mid-life crisis, and your eventual, spectacular death—is just a split second where one side gets the upper hand. The balance tips, everything goes haywire, and you have to scramble to find a brand new way to stay alive.

2 The Classical Equations of Stellar Structure

2.1 Hydrostatic Equilibrium

Consider a thin spherical shell of stellar gas at radius r , thickness dr , and cross-sectional area $A \approx 4\pi r^2$.

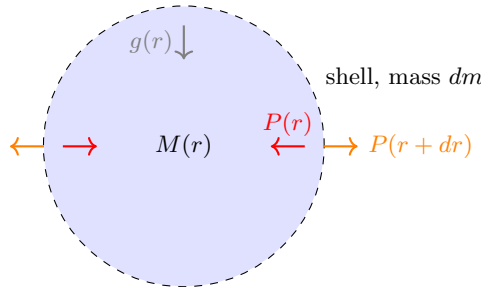


Figure 1: A thin shell inside a star. Pressure from below ($P(r)$, red) pushes outward; pressure from above ($P(r + dr)$, orange) and gravity ($g(r)$, gray) push inward.

Three forces act on this shell: pressure from below pushing it outward ($P(r)A$), pressure from above pushing it inward ($P(r + dr)A$), and its own weight, pulled inward by the gravity of all the mass $M(r)$ interior to it (by Newton's shell theorem, the mass *outside* the shell contributes no net force). The shell's mass is $dm = \rho(r)A dr$, so its weight is $\frac{GM(r)}{r^2}\rho(r)A dr$. For the star to be static (neither expanding nor collapsing), these forces must balance exactly:

$$P(r)A - P(r + dr)A - \frac{GM(r)}{r^2}\rho(r)A dr = 0.$$

Dividing through by $A dr$ and recognizing $\frac{P(r+dr)-P(r)}{dr} \rightarrow \frac{dP}{dr}$ as $dr \rightarrow 0$:

$$\boxed{\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}.$$

The pressure must always *decrease* outward (hence the minus sign) — exactly as it must, since it’s the pressure gradient, not the pressure itself, that supplies the outward force needed to hold every shell up against gravity.

2.2 Mass Conservation

The same shell, of thickness dr and density $\rho(r)$, contributes a sliver of mass $dM = \rho(r) \cdot (4\pi r^2 dr)$ to the total mass enclosed. In the differential limit,

$$\frac{dM}{dr} = 4\pi r^2 \rho(r).$$

This equation is really just bookkeeping — it says nothing more than “mass equals density times volume,” applied one thin shell at a time — but paired with hydrostatic equilibrium, it’s one of the two structural backbones that every stellar model has to satisfy simultaneously.

3 The Nuclear Engine

3.1 Mass Defect and Binding Energy

The mass of any bound nucleus is always slightly *less* than the sum of the masses of its free, separated protons and neutrons. That missing mass, the **mass defect** Δm , was converted into the energy that holds the nucleus together, via Einstein’s relation:

$$E_b = \Delta m c^2 = [Zm_p + Nm_n - M_{nucleus}] c^2.$$

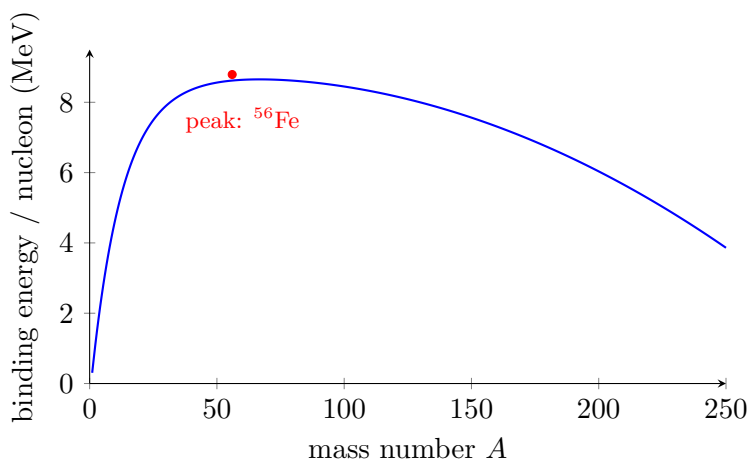


Figure 2: Binding energy per nucleon vs. mass number. It rises steeply for light nuclei, peaks near iron ($A \approx 56$), then declines slowly for heavier nuclei.

This curve is the entire reason stars can shine at all: it means **fusing light nuclei together** (climbing up the left side of the curve, toward the iron peak) releases energy, and so does **splitting very heavy nuclei apart** (fission, sliding down the right side of the curve back toward iron). Iron-56 sits at the summit — it is the most tightly bound nucleus per nucleon in existence, which is precisely why it is also nuclear *ash*: no further fusion or fission reaction starting from iron can release energy, since both directions from the peak require climbing back *up* the curve.

3.2 Quantum Tunneling and the Gamow Peak

For two nuclei to fuse, they must first get close enough for the short-range strong force to take over — but as like-charged particles, they feel a mutual Coulomb repulsion that grows without bound as they approach. Classically, stellar core temperatures ($\sim 10^7 K$) correspond to thermal kinetic energies far too small to climb over this **Coulomb barrier**. Fusion in stars only happens at all because of **quantum tunneling**: there's always some nonzero probability of a particle passing straight through a barrier it classically couldn't surmount, and that probability increases sharply with the particle's energy.

Two competing factors are therefore in play as a function of the interacting particles' energy E :

- The Maxwell-Boltzmann distribution of thermal energies falls off as $\sim e^{-E/kT}$ — higher energies are exponentially rarer.
- The tunneling probability rises sharply with energy, roughly as $\sim e^{-b/\sqrt{E}}$ (the **Gamow factor**), for a constant b set by the charges and reduced mass of the two particles.

Multiplying these together, the reaction rate at a given energy is proportional to $\exp\left(-\frac{E}{kT} - \frac{b}{\sqrt{E}}\right)$: a product of one factor that falls and one that rises, which must therefore have a maximum somewhere in between. Differentiating the exponent and setting it to zero,

$$\frac{d}{dE} \left(-\frac{E}{kT} - bE^{-1/2} \right) = -\frac{1}{kT} + \frac{b}{2}E^{-3/2} = 0 \implies \boxed{E_0 = \left(\frac{bkT}{2} \right)^{2/3}}.$$

This is the **Gamow peak** — the sweet-spot energy at which fusion reactions actually occur most often in a stellar core: too low, and particles can't tunnel through the barrier; too high, and there simply aren't enough particles with that much thermal energy to begin with.

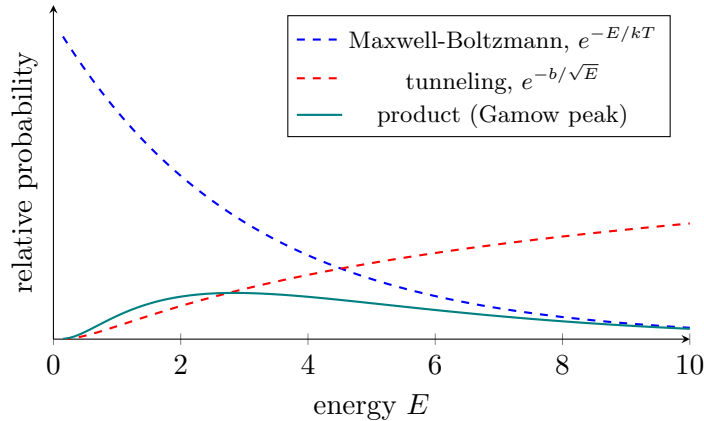


Figure 3: The Gamow peak (teal) as the product of a falling thermal distribution (blue) and a rising tunneling probability (red).

3.3 Core Fusion Tracks: PP Chain vs. CNO Cycle

Hydrogen fusion into helium proceeds via two distinct routes, depending on core temperature:

- The **proton-proton (PP) chain** fuses protons directly, step by step, into helium. Since it involves only single protons colliding (charge $Z = 1$ on each side), its Coulomb barrier is the lowest possible, and its energy generation rate scales relatively gently, roughly $\varepsilon_{pp} \propto T^4$.

- The **CNO cycle** instead uses carbon, nitrogen, and oxygen nuclei as catalysts, fusing protons onto them one at a time in a cycle that regenerates the original carbon nucleus at the end. Because the catalyst nuclei carry much higher charge ($Z = 6-8$), the Coulomb barrier is far steeper, and the Gamow-peak sensitivity to temperature becomes correspondingly extreme: $\varepsilon_{CNO} \propto T^{15-20}$.

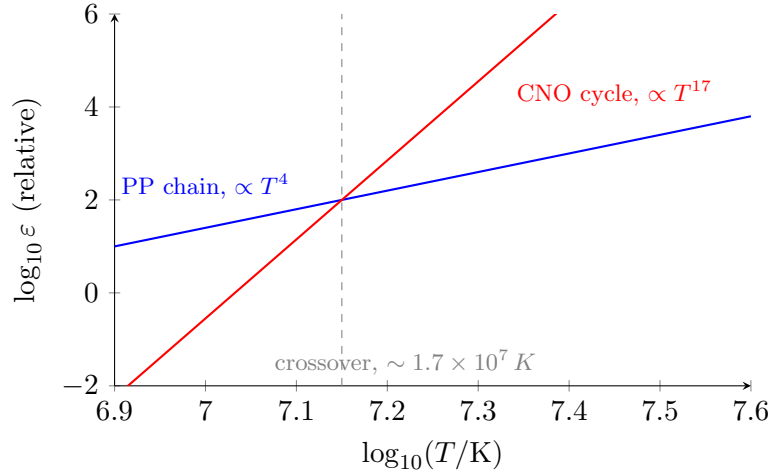


Figure 4: Energy generation rate vs. core temperature for both channels (log-log). The CNO cycle's much steeper slope means it dominates only above a sharp crossover temperature, roughly the Sun's own core temperature.

Below the crossover ($\sim 1.7 \times 10^7 K$, close to the Sun's own core temperature), the PP chain dominates; above it, in the hotter cores of more massive stars, the CNO cycle takes over almost completely, thanks to its ferociously steep temperature sensitivity.

3.4 Energy Transport: Radiative Gradient vs. the Schwarzschild Criterion

Energy generated in the core must still get out. In the **radiative diffusion** approximation, photons perform the same random walk described in Handout 4 (Section 2.1), and this random walk can be shown to carry a flux

$$F(r) = -\frac{4acT^3}{3\kappa\rho} \frac{dT}{dr},$$

where κ is the material's opacity (its resistance to photon diffusion) and a is the radiation constant. Since $L(r) = 4\pi r^2 F(r)$, solving for the temperature gradient gives

$$\left(\frac{dT}{dr} \right)_{rad} = -\frac{3\kappa\rho L(r)}{16\pi a c r^2 T^3}.$$

This is the temperature gradient the star *would* need, if radiation alone were responsible for carrying all the luminosity outward. But sometimes this required gradient is simply too steep to be stable: if a blob of gas nudged slightly upward would find itself less dense than its surroundings (and so buoyant), it keeps rising instead of sinking back — and convection takes over as a faster, bulk-fluid transport mechanism. The **Schwarzschild criterion** formalizes this: convection is triggered exactly when

$$\left| \frac{d \ln T}{d \ln P} \right|_{rad} > \left| \frac{d \ln T}{d \ln P} \right|_{ad},$$

i.e., when the radiative gradient required to carry the luminosity is steeper than the gradient a parcel of gas would follow if it rose adiabatically (without exchanging heat with its surroundings). This is exactly why the Sun is radiative in its interior (where L and κ are modest enough for radiation to keep up) but convective in its outer envelope (where falling temperature sharply increases opacity, forcing the radiative gradient to steepen past the adiabatic one).

4 Star Birth and Main Sequence Entry

4.1 The Jeans Mass

A cloud of interstellar gas collapses under its own gravity only if gravity can win before internal pressure has a chance to respond and push back. This is a race between two timescales: the **free-fall time**, $t_{ff} \sim 1/\sqrt{G\rho}$ (how long gravity takes to collapse the cloud), and the **sound-crossing time**, $t_s \sim R/c_s$ (how long a pressure wave, moving at the local sound speed $c_s = \sqrt{kT/\mu m_H}$, takes to cross the cloud and organize a pressure response).

Collapse wins when $t_{ff} < t_s$. Setting them equal defines the critical (**Jeans**) length at the boundary:

$$\frac{1}{\sqrt{G\rho}} \sim \frac{\lambda_J}{c_s} \quad \Rightarrow \quad \boxed{\lambda_J \sim \sqrt{\frac{kT}{\mu m_H G \rho}} \propto \left(\frac{T}{\rho}\right)^{1/2}}.$$

Clouds larger than λ_J can't communicate a supporting pressure response fast enough, and collapse. Converting this critical length into a critical **mass** just multiplies by the enclosed density, $M_J \sim \rho \lambda_J^3$:

$$M_J \sim \rho \left(\frac{T}{\rho}\right)^{3/2} \quad \Rightarrow \quad \boxed{M_J \propto \frac{T^{3/2}}{\rho^{1/2}}}.$$

A cloud denser than the corresponding Jeans mass for its own temperature and density is unstable and will collapse — the first step on the road to becoming a star.

4.2 The Virial Theorem

For any self-gravitating system in a steady, bound state, there's a beautifully simple relationship between its total kinetic energy K and gravitational potential energy U :

$$\boxed{2K + U = 0}.$$

This single relation has an immediate, and kinda counterintuitive, effect for a collapsing protostellar cloud. As the cloud shrinks, U becomes more negative by some amount $|\Delta U|$; the virial theorem demands that K increase by exactly *half* that amount, $\Delta K = -\Delta U/2$. The system's total energy, $E = K + U$, therefore changes by $\Delta E = \Delta K + \Delta U = \Delta U/2$ — it becomes more negative (more tightly bound) by only half of the potential energy released. The other half of the released energy goes directly into ΔK , i.e., into heating the gas. This is precisely why gravitational contraction *alone* (before any fusion begins) is enough to heat a protostar's core toward ignition temperature: half of every joule of gravitational energy released is retained as heat, while the other half is radiated away into space.

4.3 Main Sequence Lifetime Scaling

A star's main-sequence lifetime is, to order of magnitude, just its fuel supply divided by its burn rate:

$$\tau_{MS} \sim \frac{E_{fuel}}{L}.$$

The available nuclear fuel is proportional to the star's total mass, $E_{fuel} \propto M$, so

$$\tau_{MS} \propto \frac{M}{L}.$$

Empirically (and from the steep core-temperature sensitivity of nuclear burning derived in Section 2.4), stellar luminosity climbs steeply with mass, roughly as $L \propto M^{3.5}$. Substituting,

$$\tau_{MS} \propto \frac{M}{M^{3.5}} \implies \boxed{\tau_{MS} \propto M^{-2.5}}.$$

Massive stars, despite having far more fuel in absolute terms, burn through it so much faster that they live dramatically *shorter* lives — a $20M_{\odot}$ star lives roughly $20^{-2.5} \approx 0.06\%$ as long as the Sun, only a few million years, while a red dwarf a tenth the Sun's mass can outlive the current age of the universe many times over.

5 Post-Main Sequence Pathways and Stellar Death

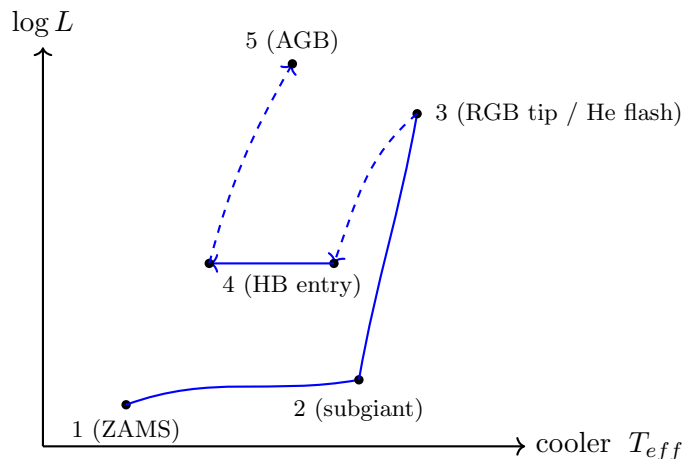


Figure 5: Schematic post-main-sequence track for a Sun-like star on the H-R diagram (temperature decreasing to the right, as is conventional): main sequence (1) → subgiant branch (2) → red giant branch ascent terminating at the helium flash (3) → rapid contraction onto the horizontal branch entry (4) → horizontal branch evolution followed by ascent up the asymptotic giant branch (5).

5.1 Shell Burning, the Subgiant Branch, and the Red Giant Branch

Once core hydrogen is exhausted, the core — now almost pure helium, with no fusion to support it — has lost its pressure source and begins to contract under its own gravity. That contraction releases gravitational energy (exactly per the virial argument of Section 4.2), and much of that energy heats the hydrogen-rich layer immediately *surrounding* the core hot enough to ignite: **hydrogen shell**

burning begins. This sets up the **mirror principle**, a recurring theme in stellar evolution: as the inert core continues contracting, the shell burning above it drives the star's outer envelope to expand and cool, moving the star up and to the right on the H-R diagram, first onto the **subgiant branch**, and then, as the envelope becomes fully convective, up the steep, nearly vertical **red giant branch (RGB)**.

5.2 Helium Flash and the Horizontal Branch

As the core continues to contract, it eventually becomes so dense that **electron degeneracy pressure** (Section 5.1) takes over as its main support — and degeneracy pressure, crucially, does not depend on temperature at all. When the core finally reaches the $\sim 10^8 K$ needed to ignite helium fusion (the triple-alpha process), the resulting energy release raises the temperature further, which would normally make the core expand and cool, throttling the reaction — but a degenerate core doesn't expand in response to heating. The result is a runaway: the **helium flash**, an explosive spike in the fusion rate (briefly rivaling the entire galaxy's luminosity, though hidden deep inside the star and never seen directly) that only ends once the energy release finally becomes large enough to lift the degeneracy itself, letting the core expand and settle into stable, non-degenerate helium burning. The star then finds a new, quieter equilibrium on the **horizontal branch**, burning helium in its core and hydrogen in a surrounding shell simultaneously.

5.3 Asymptotic Giant Branch (AGB) Thermal Pulses

Once core helium is also exhausted (built up into an inert carbon-oxygen core), the star repeats the same mirror-principle trick one level further out, developing *two* concentric burning shells — helium below, hydrogen above — on the **asymptotic giant branch**. These two shells do not burn steadily; they alternate in a cycle of **thermal pulses**: the helium shell periodically ignites in its own localized flash (for the same degeneracy-driven runaway reason as Section 4.2), briefly overwhelms and shuts off the hydrogen shell above it, then subsides, letting hydrogen burning resume until enough helium ash accumulates to trigger the next pulse. Each pulse drives a convective dredge-up event that can carry freshly synthesized elements from deep in the star up to its surface, which is how AGB stars enrich the interstellar medium with carbon and other processed material.

5.4 Mass Loss Dynamics (Qualitative)

Throughout the giant branches, a star's enormously extended, weakly-bound outer envelope is progressively lost as a **stellar wind**, driven by radiation pressure (Handout 4, Section 2.5) acting on dust grains and molecules forming in the cool, extended atmosphere; empirically, mass-loss rate scales steeply with luminosity and inversely with effective temperature, so it's the giant phases — not the main sequence — where a star sheds the overwhelming majority of its mass. By the end of the AGB, the entire envelope can be ejected in a comparatively short-lived, glowing shell called a **planetary nebula** (a historical misnomer — it has nothing to do with planets), exposing the hot, inert carbon-oxygen core beneath: a white dwarf.

6 Quantum Degeneracy and Compact End-States

6.1 Electron Degeneracy Pressure

When electrons are packed densely enough, the Pauli exclusion principle forbids any two of them from occupying the same quantum state — and combined with the Heisenberg uncertainty principle,

this forces most electrons to have large momenta, whether or not the gas is hot. Confining an electron to a volume of size $\sim 1/n$ (where n is the number density) requires a position uncertainty $\Delta x \sim n^{-1/3}$, and so a minimum momentum uncertainty

$$\Delta p \sim \frac{\hbar}{\Delta x} \sim \hbar n^{1/3}.$$

Treating this as the characteristic (Fermi) momentum of the most energetic electrons, the non-relativistic kinetic energy per electron is $\sim p^2/2m_e \sim \hbar^2 n^{2/3}/m_e$, and the degeneracy pressure — roughly, number density times energy per particle — becomes

$$P_{deg} \sim n \cdot \frac{\hbar^2 n^{2/3}}{m_e} = \frac{\hbar^2}{m_e} n^{5/3}.$$

Since $n \propto \rho$,

$$P_{deg} \propto \rho^{5/3}.$$

The crucial feature of this pressure is that it depends *only* on density, not temperature at all — a degenerate gas can be squeezed to arbitrarily high pressure without needing to be hot, which is exactly what supports white dwarfs (and, transiently, the helium-flash core of Section 4.2) even as they cool.

6.2 The Chandrasekhar Limit

As a white dwarf's mass increases, its electrons are squeezed to ever-higher densities, and eventually their Fermi momentum becomes large enough that they move at relativistic speeds. In this **ultra-relativistic** regime, the energy per electron becomes $\sim pc \sim \hbar c n^{1/3}$ instead of $p^2/2m_e$, so the pressure scaling changes:

$$P_{deg,rel} \sim n \cdot \hbar c n^{1/3} = \hbar c n^{4/3} \propto \rho^{4/3}.$$

Meanwhile, the pressure gravity *demands* at the center of a self-gravitating sphere scales (from hydrostatic equilibrium, Section 1.1, order-of-magnitude) as $P_{grav} \sim GM^2/R^4$, and since $R \propto (M/\rho)^{1/3}$ for roughly constant density, this becomes $P_{grav} \sim GM^{2/3}\rho^{4/3}$.

Both the relativistic degenerate pressure and the pressure gravity demands scale identically with $\rho^{4/3}$ — meaning that above some critical mass, the $\rho^{4/3}$ terms cancel out of the equilibrium condition entirely, and the two sides can *only* balance for one specific value of M , not for a whole family of stable radii. Working through the full coefficients (using detailed relativistic degenerate-gas physics beyond this order-of-magnitude sketch) yields

$$M_{Ch} \approx 1.4 M_{\odot}.$$

Below this mass, an increase in pressure from further compression can still out-pace gravity's own demands as R shrinks, so a stable radius always exists. Above it, gravity wins at every radius, and no stable degenerate configuration exists at all — the star must collapse further, toward a neutron star or black hole.

6.3 Neutron Stars and Pulsars

If the collapsing core exceeds the Chandrasekhar limit, its density climbs so high that inverse beta decay ($p^+ + e^- \rightarrow n + \nu_e$) becomes favorable, converting the core almost entirely into neutrons. This new gas of neutrons can itself become degenerate, supported by **neutron degeneracy pressure**

— structurally identical to Section 5.1’s derivation, just with m_e replaced by the much larger m_n , allowing neutron stars to support far higher densities before their own analogous mass limit (the Tolman-Oppenheimer-Volkoff limit, $\sim 2\text{--}3 M_\odot$, though its exact value depends on the still-uncertain nuclear equation of state at these densities).

Two further effects fall directly out of conservation laws during the collapse itself:

- **Rotation:** angular momentum $L = I\omega$ is conserved as the core collapses from roughly the size of the Sun down to a radius of order 10 km . Since $I \propto R^2$, an enormous decrease in R forces a correspondingly enormous increase in ω — this is exactly why neutron stars are observed to spin at rates up to hundreds of times per second. A **pulsar** is such a rapidly rotating, strongly magnetized neutron star whose beamed radio emission sweeps past Earth once per rotation, like a lighthouse.
- **Magnetic field:** magnetic flux $\Phi \sim BR^2$ is likewise conserved during collapse, so the same dramatic shrinkage in R forces a dramatic increase in B , producing the famously enormous magnetic fields ($\sim 10^8\text{--}10^{15}$ Gauss) observed in neutron stars.

6.4 Black Holes

If even neutron degeneracy pressure is overwhelmed, no known force can halt further collapse, and the object shrinks without limit. The boundary beyond which nothing — not even light — can escape is the **event horizon**, at the **Schwarzschild radius**. A fully rigorous derivation requires general relativity, but a naive Newtonian argument — setting the escape velocity equal to the speed of light — happens to reproduce the exact right answer:

$$\frac{1}{2}mv_{esc}^2 = \frac{GMm}{r} \implies v_{esc} = \sqrt{\frac{2GM}{r}} \xrightarrow{v_{esc} \rightarrow c} \boxed{R_s = \frac{2GM}{c^2}}.$$

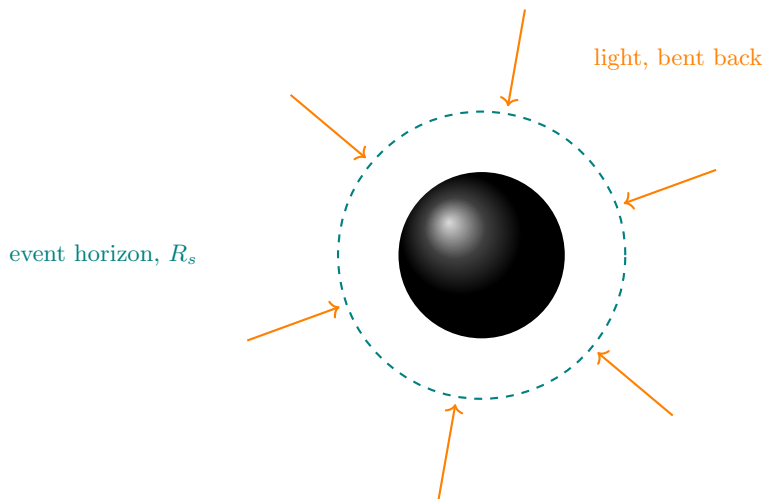


Figure 6: The Schwarzschild radius marks the point of no return: within it, not even light can climb back out.

For a solar-mass object, $R_s \approx 3\text{ km}$ — meaning a black hole of the Sun’s own mass would need the Sun’s entire mass compressed inside a sphere smaller than most cities.

7 Summary Sheet

Concept	Formula
Hydrostatic equilibrium	$dP/dr = -GM(r)\rho(r)/r^2$
Mass conservation	$dM/dr = 4\pi r^2 \rho(r)$
Binding energy	$E_b = \Delta m c^2$
Gamow peak energy	$E_0 = (bkT/2)^{2/3}$
PP chain / CNO cycle scaling	$\varepsilon_{pp} \propto T^4, \varepsilon_{CNO} \propto T^{15-20}$
Radiative temperature gradient	$(dT/dr)_{rad} = -3\kappa\rho L/(16\pi acr^2 T^3)$
Schwarzschild (convective) criterion	$ d \ln T / d \ln P _{rad} > d \ln T / d \ln P _{ad}$
Jeans mass	$M_J \propto T^{3/2} / \rho^{1/2}$
Virial theorem	$2K + U = 0$
Main-sequence lifetime	$\tau_{MS} \propto M^{-2.5}$ (using $L \propto M^{3.5}$)
Non-relativistic electron degeneracy pressure	$P_{deg} \propto \rho^{5/3}$
Chandrasekhar limit	$M_{Ch} \approx 1.4 M_\odot$
Schwarzschild radius	$R_s = 2GM/c^2$